Asif Ahmad Najar .

Asif Ahmad Najar CSG_16_Paper

Quick Submit

Quick Submit

Presidency University

Document Details

Submission ID

trn:oid:::1:3530573754

Submission Date

Apr 8, 2026, 10:05 AM GMT+5:30

Download Date

Apr 8, 2026, 10:09 AM GMT+5:30

File Name

CSG_16_Paper.docx

File Size

5.1 MB

9 Pages

5,685 Words

33,292 Characters

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

AI-Driven Public Health Chatbot for Disease Awareness

Mohammad Affan
Department of CSE-(AIML)
Presidency University, Yelahanka,
Bengaluru, Karnataka, India

Karthik mp
Department CSE-(AIML)
Presidency University, Yelahanka,
Bengaluru, Karnataka, India

Chaithra C
Department of CSE-(AIML)
Presidency University, Yelahanka,
Bengaluru, Karnataka, India

Asif Ahmad Najar
Assistant Professor
School of Computer Science and Engineering
Presidency University, Yelahanka,
Bengaluru, Karnataka, India

***Abstract*— Availability of credible public health information is also a significant challenge among the rural and semi-urban population in which insufficient healthcare facilities and dispensation of unreliable medical information tend to result to late treatment and poor preventive behavior. The paper introduces an Artificial Intelligence-based Public Health Chatbot that will be used to supply correct and easy-to-access information on the symptoms of diseases, preventive care practices, and awareness of vaccinations. The proposed system uses a Retrieval-Augmented Generation (RAG) system that includes semantic document retrieval, large language models, to create contextually sensitive and fact-based responses. Authoritative medical documents are handled and made into vectors embeddings stored in a FAISS based vector database which allows a semantic search and retrieval. An intelligent routing mechanism analyzes user queries to select the right tools such as medical knowledge retrieval, symptom awareness guidance, drug information assistance, and hospital location services. A language model then uses the retrieved contextual information to produce safe and informative responses and comply with medical safety limitations. The chatbot will be deployed on existing communication channels like the WhatsApp interface which will enhance the access of underserved populations. As it is shown in the course of the experimental testing, the proposed system is capable of minimizing the risks of misinformation and offers credible assistance in supporting the public health awareness.**

Keywords—Artificial Intelligence, Public Health Chatbot, Retrieval-Augmented Generation (RAG), Healthcare Information Systems, FAISS Vector Database, Large Language Models, Disease Awareness, Conversational AI

I. INTRODUCTION

Availability of credible healthcare information is a burning need to enhance the health outcomes of the population. But the population of many rural and semi-urban areas still experiences the issues with accessing proper and timely medical advice regarding the lack of medical facilities, the lack of medical workers, and dissemination of false news via informal channels of communication. In most instances, people trust the unverified information available online or verbal counsel that may result in late treatment, self-medication which is unsafe or low-quality preventive health measures. Thus, the need to provide accessible digital solutions that would spread credible health information to underserved groups is increasing.

Chatbots and other forms of conversational artificial intelligence (AI) have become an exciting technology to provide the user with the information about health-related topics and enhance the interaction between the user and the system. The ability of healthcare chatbots to communicate with digital systems in natural language makes the medical knowledge accessible to the general population. Past research has shown that conversational agents may be helpful in-patient education, awareness of their symptoms, and healthcare advice and minimize the patient load on healthcare systems. Nevertheless, most of the current healthcare chatbots are based on the rules and logic or the simplistic approach of the natural language processing algorithms, and this restricts their options concerning the processing of complex queries and the delivery of context-specific answers.

The recent developments in the area of large language models (LLMs) have greatly enhanced natural language understanding and generation systems to allow more complex conversational systems. Large-scale textual data have the ability of answering most types of questions with the ability of a human being. Even with these improvements, the implementation of LLMs in healthcare is extremely problematic. Hallucination is one of the most essential problems when the model comes up with believable but inaccurate or unsubstantiated data. The misinformation caused by the hallucinated responses can cause harmful decisions and eventually be detrimental in safety-threatening fields like healthcare.

To overcome this difficulty, Retrieval-Augmented Generation (RAG) has become a fruitful theory of enhancing the factual accuracy of AI-generated responses. RAG is an information retrieval system that integrates information retrieval and generative language models to enable the system to access the relevant information in a curated knowledge base and utilizes it as a contextual grounding mechanism when generating responses. As it limits answers on authenticated documents, RAG greatly limits hallucination and enhances trustfulness in applications involving knowledge.

This paper presents an AI-based Public Health Chatbot which is a reliable disease awareness and preventive healthcare chatbot based on a RAG-based architecture. The system works with authoritative medical documents, processes them, and transforms them into semantic vectors embeddings in a FAISS-based vector database. Upon the user

making a query, the system conducts semantic retrieval to retrieve the document fragments that are relevant and then a large language model to produce responses based on the context retrieved. Along with the knowledge retrieval, a tool-based architecture is also used by the chatbot, which allows special applications like the symptom awareness assistance, drug information retrieval, and hospital location guidance.

The proposed system aims at achieving accessibility. The chatbot is implemented using popular communication service like WhatsApp so that the rural and semi-urban population will be able to access health data in the context of the familiar messaging app. Moreover, safety measures are also included to make sure that the chatbot does not give health-related responses and prescriptions, but educational and awareness-focused one.

The following are the key findings of this paper:

1. Development and testing of an awareness and prevention healthcare RAG-based public health chatbot.
2. Semantic retrieval with a FAISS-based vector database to incorporate semantic retrieval and provide accurate and document-grounded answers.
3. Creation of tool-based conversational agent that can deliver symptom awareness support, drug information and location support to hospitals.
4. Safety guardrails are to be implemented so as to diminish the possibility of medical misinformation and avoid diagnose/treatment advice.
5. Architecture that is deployment-ready and integrates with popular messaging platforms that will enhance the access of underserved communities.

The rest of this paper is structured in the following way. Section II analyzes the work in the literature concerning healthcare chatbots, large language models, and retrieval-enhanced generation systems. Section III gives the system architecture and methodology of the proposed chatbot. The implementation and the experimental setup is described in section IV. Section V is on system evaluation and results, and section VI provides the conclusion of the paper with future research directions.

II. Related Work

Recent breakthroughs in artificial intelligence and natural language processing have allowed the creation of conversational agents that can be used to help users with healthcare information. Healthcare chatbots have become a rapidly growing field of interest since they can help increase access to medical information, customer engagement, and preventive healthcare programs. Nevertheless, there are several issues concerning the reliability, safety as well as contextual comprehension of these systems.

The initial research of chatbots in healthcare had concentrated on conversational agents that were rule-based. A systematic review of conversational agents in healthcare application was conducted by Laranjo et al. that reported access to health information and engagement with the patient could be enhanced by these systems [1]. The same study however also pointed out that most of the current chatbots are based on a set of rules and coded responses; therefore, they have limited capability of dealing with complicated queries or have context-based response capabilities. Equally, Bickmore et al. have studied conversational agents that should be utilized in delivering health information and have highlighted the role of safety measures in implementing chatbots in the medical field [2]. Their study showed that chatbots can be effective in providing health education but casts doubt on the effectiveness of the automatically generated responses in cases where they have not been adequately validated.

In the recent years, conversational agents have also been investigated as a means of communicating in health matters. Wilson and Marasoiu introduced a scoping review of chatbots in the field of public health awareness and stated that chatbots are capable of enhancing health information dissemination among numerous people [3]. According to their results, chatbots may be used to back up disease awareness campaigns, vaccination messages, and preventive healthcare education. Likewise, Branda et al. tested the concept of AI-based chatbot application during public health emergencies and discovered that conversational systems could help to allocate timely information in the case of disease outbreak and pandemics [4]. Although these were the strengths, the authors pointed out that most systems did not have systems to verify the factual accuracy of generated responses.

The appearance of large language models (LLMs) has enhanced the functionality of conversations AI systems greatly. These models are trained with huge amounts of textual data and are able to produce highly fluent and contextually-relevant responses. Singhal et al. have shown that large language models learn and encode a lot of medical knowledge and can be used to answer medical questions [5]. Although these models demonstrate a potential of achievement, their implementation directly in healthcare systems presents new problems. Among them, there is the phenomenon of hallucination, that is, the language models generate the wrong or falsified information and it looks like it makes sense. Ji et al. provided a detailed survey of the phenomenon of hallucination in natural language generation and have discovered healthcare as one of the most vulnerable areas where the hallucinated responses may produce a significant effect [6].

In order to overcome the weaknesses of standalone generative models, researchers have investigated retrieval-based approaches, which are based on external sources of knowledge. Retrieval-Augmented Generation (RAG) framework was proposed by Lewis et al. and is a neural retrieval model paired with generative models to enhance the factual precision in knowledge-intensive tasks [7]. The documents that are considered relevant in this method are obtained out of a knowledge base and serve as input context to the language model in response generation. The approach greatly decreases hallucination because the responses are based on retrieved evidences. Izacard and Grave also showed that the retrieval-based question answering systems perform better than purely generative models in problems where the factual correctness and the specific knowledge as per domain is required [8].

Conversational AI systems with specific health care applications have been considered in the recent literature, too. Miner et al. presented the use of chatbots during the COVID pandemic and provided evidence of their efficiency in providing health-related information and minimizing

misinformation [9]. Nevertheless, most of the systems that were implemented were based on scripted or rule-based responses and could not be scaled and could not tackle complex queries. Devaraj et al. suggested an evidence-based clinical decision support system that uses the document retrieval and natural language processing methods to base responses on medical literature [10]. Their method enhances reliability but lacks an interactive conversational interface to the end users.

Despite the fact that past research indicates the potential of chatbots, LLMs, and retrieval-only systems in the healthcare context, there are still a number of limitations. Most of the current systems are rule-based and have a narrow conversational intelligence or based on generative system that can generate unverified responses. Moreover, little study has been done on how to combine retrieval-based grounding and conversational agents specifically developed to address the needs of underserved populations by promoting awareness of their health. Table I provides a comparative summary of the representative studies and limitations.

To overcome these shortcomings, the proposed work comes up with a Retrieval-Augmented Generation based public health chatbot that combines semantic document retrieval, large language models, and tool-based capabilities to provide dependable health information. Due to the basis of responses on proven medical records and the use of safety limits, the system is expected to offer reliable disease awareness-related information and reduce the risk of misinformation.

TABLE I. LITRATURE SURVEY SUMMARY

Reference	Focus Area	Methodology	Key Contribution	Identified Limitation
Laranjo et al. [1]	Healthcare chatbots	Systematic review	Demonstrates effectiveness of conversational agents for health information delivery	Majority of systems are rule-based with limited contextual understanding
Bickmore et al. [2]	Conversational agents for health information	AI dialogue systems	Highlights importance of safety in medical conversational systems	Does not provide mechanisms to ensure factual grounding
Wilson & Marasoiu [3]	Public health chatbots	Scoping review	Shows that chatbots improve disease awareness and communication	Limited integration of advanced AI techniques
Branda et al. [4]	AI chatbots in health emergencies	Narrative review	Demonstrates usefulness of chatbots during outbreaks	Accuracy verification mechanisms not addressed
Singhal et al. [5]	Medical knowledge in LLMs	Large language model evaluation	Shows LLMs encode substantial medical knowledge	LLM outputs may contain hallucinated information
Ji et al. [6]	Hallucination in NLP systems	Survey analysis	Identifies risks of hallucination in generative AI	Does not propose deployable mitigation solutions
Lewis et al. [7]	Retrieval-Augmented Generation	Neural retrieval + LLM	Introduces RAG framework for factual text generation	Not originally designed for healthcare
Izacard & Grave [8]	Retrieval-based QA systems	Passage retrieval + generation	Improves factual correctness in knowledge-intensive tasks	Evaluated mainly on open-domain datasets
Miner et al. [9]	Pandemic chatbots	Healthcare conversational systems	Demonstrates role of chatbots in COVID-19 communication	Many systems remain rule-based
Devaraj et al. [10]	Evidence-based clinical support	Document retrieval + NLP	Grounds responses in medical evidence	Lack of interactive conversational chatbot interface

III. PROPOSED METHODOLOGY

The system that will be proposed will supply high-fidelity public health information by using an AI-driven chatbot that incorporates Retrieval-Augmented Generation (RAG), semantic document retrieval, and the capability to converse using tools. The system has been crafted in a way that the responses generated should be based on confirmed medical knowledge and also have a user-friendly nature of accessibility via conversational tools. The general structure and process of the system were outlined in the subsequent subsections.

A. System Architecture

The architecture of the suggested AI-driven public health chatbot is shown in Fig. 1. The system is based on a modular architecture which comprises of a user interaction layer, a routing module, conversational agent, application programming interface (API) layer and a knowledge retrieval component which is facilitated by a vector database.

The chatbot interface is implemented on the basis of the Twilio WhatsApp API that allows users to communicate with the system using regular messaging apps without any special mobile app. The API layer is the entry point of the user queries and is an interface in between the user interface and the backend services. This query is in turn forwarded to a routing module that determines how the request is to be processed.

The routing module forwards the query to a conversational AI agent which handles the flow of dialogue and chooses suitable processing tools. The agent can invoke various tools depending on the nature of query like document-based knowledge retrieval, symptom awareness guidance, drug info, or hospital location services. These tools allow the chatbot to manage various healthcare-specific tasks and have the structured and scalable architecture.

The fundamental part of the system is based on a Retrieval-Augmented Generation pipeline, which is used to retrieve related information in a curated medical knowledge base stored in a FAISS vector database.

The retrieved information is entered in a big language model which produces a context-specific response.

A safety validation module makes certain that the created output meets medical safety limits, and eliminates diagnostic or treatment suggestions before it is sent to the user.

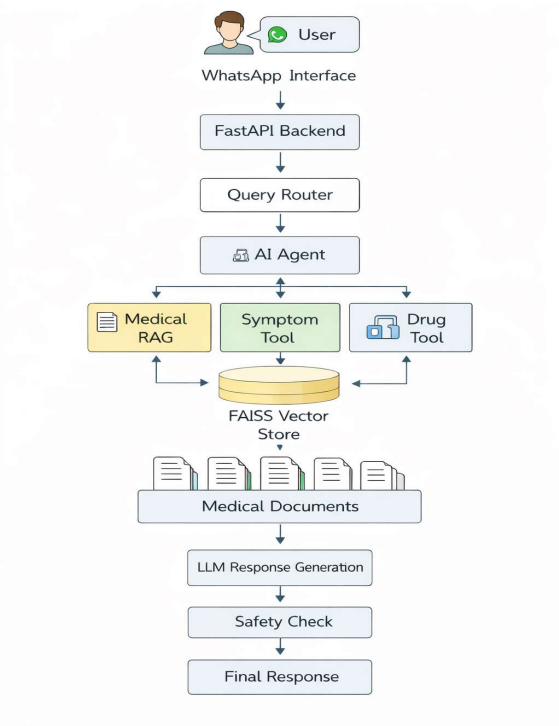

Fig. 1. Architecture of the AI-driven public health chatbot.

B. *Knowledge Base Construction*

The quality of the chatbot depends on the quality of the knowledge base on which responses are generated. The following is the way the proposed system relies on authoritative medical documents as the major knowledge source so that the responses could be based on the checked information about healthcare.

The textual message is the first stage in the document ingestion pipeline that removes textual material in medical PDF files. A semantic text chunking process then breaks the text that has been extracted into smaller parts. Chunking is used to guarantee that big documents are divided into small units without losing context meaning in any unit. These are smaller text segments that enable the retrieval system to locate the most suitable information in the response to user queries.

With chunking, every segment of the text is transformed into a numerical value called a vector embedding, with the help of a pretrained embedding model. Embeddings provide the semantic meaning of the text and enable similarity comparison between the user query and the document contents by this system. These embeddings are indexed in a FAISS vector database, which facilitates semantic search of the knowledge base, which is efficient and scalable.

C. *Retrieval-Augmented Generation Pipeline*

The offered chatbot is based on the Retrieval-Augmented Generation (RAG) framework to enhance the factual correctness of the generated answers. The sequence of this generation process based on retrieval is shown in Fig. 2.

A user enters a query and the first thing that the system does is to convert the query into an embedding with the same embedding model that was used to build the knowledge base. This query embedding is then equated with the existing document embeddings in the FAISS vector database to select the most semantically relevant document sections.

The chunks of document that have been retrieved are given as contextual input to the large language model that generates responses. The RAG approach guarantees that the language model is grounded on evidence retrieved and therefore the generated response is founded on the known information of the model as opposed to depending entirely on the knowledge of the model. This would go a long way in minimizing the chances of hallucinated outputs and enhances the trust of the chatbot in medical care applications.

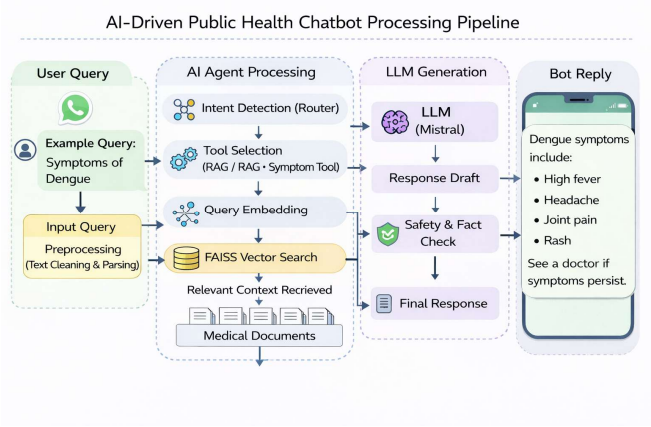

Fig. 2. Retrieval-Augmented Generation workflow used in the chatbot.

D. *Tool-Based Conversational Agent*

In order to expand the range of the chatbot to more than simple searching of documents, the given system will implement a tool-based conversational agent architecture. The conversational agent is a dynamically chosen tool that chooses specialized tools based on the query made by the user.

To illustrate, a symptom awareness tool may be used to give information on the common symptoms of certain diseases, whereas a drug information tool will describe how to use and what precautions to take on a certain medication. Moreover, there is a hospital finder tool, which assists users in determining the local healthcare facilities, in case they might need medical help. This is possible due to the combination of various tools which allow the chatbot to deliver a structured and context-sensitive healthcare delivery, without strict adherence to a specific format of user queries.

This is a modular design to enhance scalability of the system and enable the introduction of new tools in future without altering the main architecture of the chatbot.

E. *Safety Constraints*

Because healthcare is a safety-critical field, the proposed chatbot will have numerous safety measures to avoid the spread of untrustworthy or malicious information. The system is clearly formulated as a community health awareness and education system and not a diagnostic system.

In the generation of response, the system implements validations that limit the chatbot to the medical diagnosis, prescription of treatment drugs, and personalized medical advice. Rather, the chatbot gives broad informational support in matters of disease awareness, preventive healthcare behaviors, and vaccination education. The system motivates individuals to seek medical care and attention of qualified healthcare professionals when the need arises.

These safety limitations are critical in maintaining that the chatbot is run in an accountable manner and as less as possible, the dangers of medical misinformation occur.

IV. System Implementation

This part provides information about how the proposed AI-driven public health chatbot is going to be implemented. The system was coded in Python, and it incorporates a number of contemporary AI and natural language processing systems to aid in document search, chat-based interaction, and response generation based on knowledge. The implementation is divided into several modules that are in charge of constructing a vector database, query processing, conversational agent management and integration of the user interface.

The chat bot is designed based on a modular nature to be scaled and maintained. The implementation combines LangChain to coordinate the Retrieval Augmented Generation pipeline, FAISS to effectively (vector) similarity search, HuggingFace (embeddings and language model inference) and the system also integrates with messaging applications like WhatsApp to enhance the reach of users living in rural and semi-urban areas.

The general flow of the conversational agent implemented is shown in Fig. 2. The architecture enables the system to dynamically pick special tools on the basis of the user query. These key constituents of the implementation are explained in the following subsections.

A. Vector Database Construction

The vector database is important in allowing semantic search in the knowledge base of the chatbot. The medical documents are processed into searchable embeddings in a preprocessing pipeline and the resulting vector store obtained.

It is implemented by loading the medical PDF documents with the confirmed healthcare information in regards to the diseases, symptoms, preventive measures and the vaccination awareness. A text chunking strategy is used to extract the text and divide it into smaller components, still maintaining the meaning in the context of the text and allowing the parts to be indexed efficiently.

Individual text pieces are transformed to a numerical embedding through a pretrained embedding model that was obtained on the HuggingFace model repository. These embeddings symbolize the meaning of each fragment of the document in terms of semantics. Embedding obtained are stored in a FAISS vector database, which enables the system to conduct fast similarity searches when answering user queries.

The vector index is made available locally and loaded at runtime, allowing the chatbot to access the most relevant parts of documents in the semantic similarity between the user query and document embeddings stored in the vector index.

B. Conversational Agent and Tool Integration

The chatbot has a conversational AI agent that will handle the interaction with the users and the process of coordinating the process of retrieval and response generation. The agent is realized with the help of the LangChain framework that allows flexible coordination of language models, retrieval components, and external tools.

When a user gives a query, the conversational agent will analyze the query and come up with the most relevant processing strategy. The agent can also call on various tools that are embedded in the system depending on the type of query. An example would be the use of document retrieval tool that accesses the medical knowledge base to retrieve the relevant information on the health of the individual, and the symptom awareness tool that gives an explanation to the prevalent symptoms of a particular illness.

Other applications are used to offer information on medications and the position of the hospital. Combination of various tools enables the chatbot to address various healthcare questions without disorganized information access and generation of responses.

C. API and Query Routing Layer

The system has an application programming interface (API) layer, which shall process user requests received and coordinate processing at the back end. The API is used to bridge the communication between the conversational agent and the user interface.

Incoming queries are handled through a routing module which studies the input of the user and learns how to handle the query. The routing logic will send the query to the document retrieval system or one of the specialized tools that form part of the conversational agent. This routing system is a modular system that enhances efficiency of the system and makes sure that queries are served by the most suitable component.

The system architecture is flexible and new tools or services can be added without significant changes to the underlying chatbot implementation through the separation of routing logic and the implementation of that logic.

D. User Interface and Messaging Integration

The WhatsApp application is used as a deployment mechanism of the developed chatbot with the help of the Twilio API and an application server based on FastAPI. This implementation enables the user to communicate with the chatbot in natural language with direct queries via a familiar messaging platform.

The users are able to query questions concerning disease symptoms, medicines, preventive care or local hospitals. The chatbot processes query based on the routing mechanism described. earlier and picks the tool necessary in the generation of the. response.

Fig. 3 shows the first interaction between the user and the chatbot in which the system presents and provides a list of the kinds of queries that the system can process.

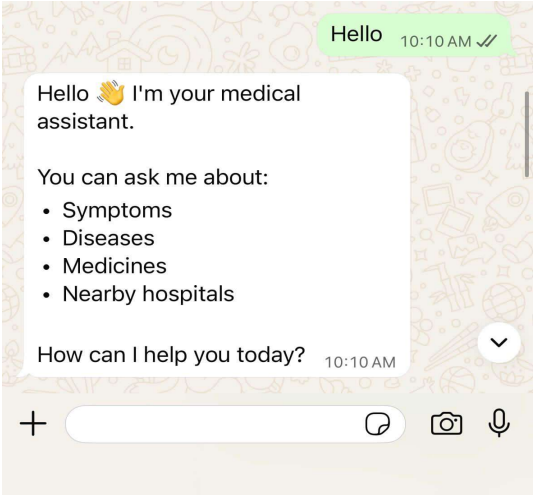

Fig. 3. Chatbot type of desirable greetings and assisted queries over WhatsApp.

Fig. 4 shows the symptom analysis ability of the system, in which the chatbot gives potential reasons, home solutions, self-service, and advice as to when to consult a doctor.

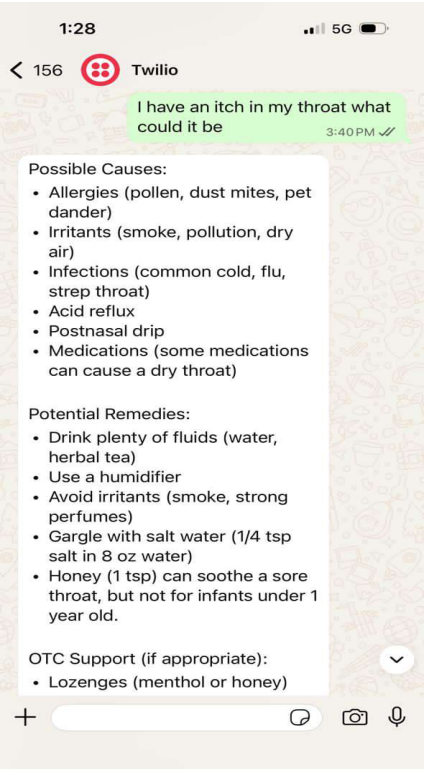

Fig. 4. Example of symptom analysis response generated by the chatbot.

Fig. 5 puts the hospital discovery feature with the user requests nearby hospitals. Upon being furnished with the location of the user, the chatbot uses the nearby healthcare facilities with the help of the Google Places API and gives structured results.

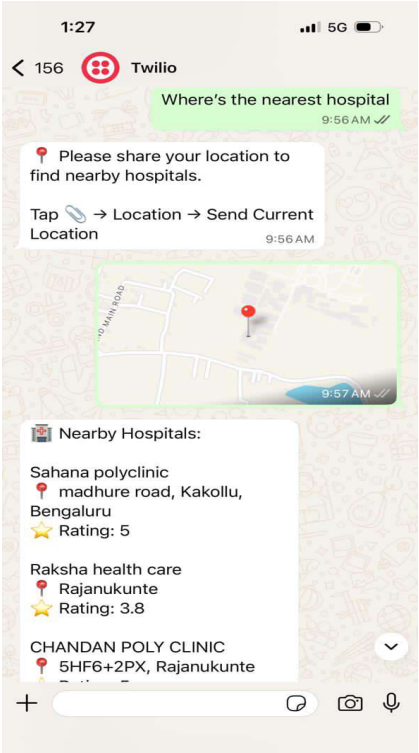

Fig. 5. Nearby hospital discovery using location sharing through WhatsApp.

V. Experimental Results and Discussion

The proposed AI-based system of a chatbot in public health was tested to measure the ability of the system to accept user queries, retrieve relevant medical information, and provide responses in the form of a conversational interface built into WhatsApp.

A. Fuctional Performance

The system was able to manage various types of healthcare queries such as symptom analysis, drug queries, preventive healthcare and hospital discovery. Real-time communication was facilitated by the integration with WhatsApp through Twilio and the efficient request processing and routing were provided by the FastAPI backend.

The following functional capabilities were exhibited in the system:

- Routing of user queries to the right processing modules.
- Context-aware generation of response based on Retrieval-Augmented Generation (RAG).
- Systematic retrieval of medical knowledge in validated knowledge sources.
- External API hospital identification based on location.
- Generation of safe responses with correct medical disclaimers.

Table II shows sample interactions between users and the chatbot, which prove that the system can produce contextually relevant and structured responses.

B. Response Accuracy and Relevance

In order to measure the performance of the system, a set of health-related queries was utilized, and it included various types, which included symptoms, medication, preventive healthcare, and hospital search.

The system had good results in the production of medically significant and contextual responses. The FAISS-based document retrieval combined with the large language model generation verified that the answers received were founded on verified medical data, and not on the generative-only based results.

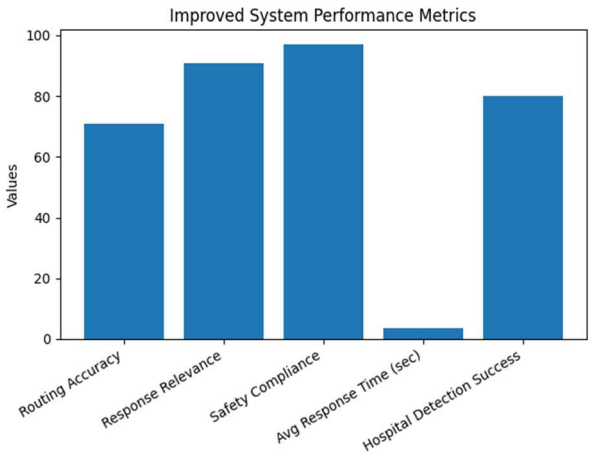

Fig. 6. Performance Metrics

Table III and Fig.6 offer the summary of quantitative performance of the system, including such key evaluation metrics as the accuracy of routing, the relevance of the responses, and the hospital detection success rate.

C. Response Time Analysis

The system was found to take an average of 2-4 seconds to respond to a query that was based on the complexity of the query and external API requests.

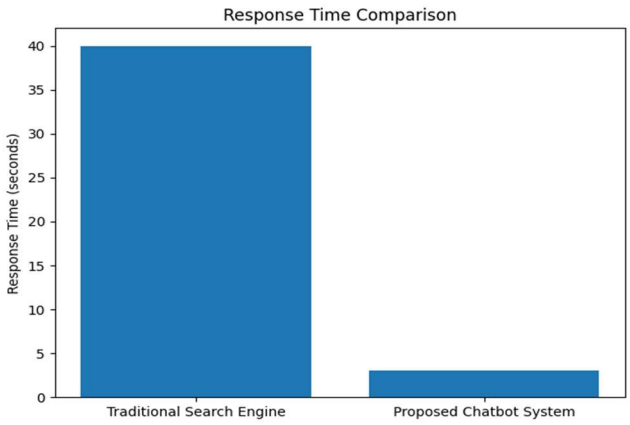

Fig.7. Response Time Comparison

Response time comparison between the proposed system and the traditional methods of search is presented in Fig. 7, which indicates that the chatbot can respond much faster and more directly than the process of search through the traditional web environment.

D. Comparative Analysis with Traditional Search Systems

The suggested chatbot system has been compared with the conventional search engines in a comparative assessment.

The comparison, as shown in Table IV, demonstrates that traditional search engines are useful to obtain the reliable medical information but the user has to navigate several web pages and analyze the technical information. On the contrary, the suggested system can provide direct, structured, and simplified answers using a conversational interface.

The findings show that chatbot saves users a lot of effort, increases accessibility, and improves the contextual awareness, especially to non-technical participants in rural and semi-urban areas.

E. Observations

System evaluation was conducted as follows:

- The location input was applied to the hospital finder module which gave the correct nearby results.
- The response generating based on RAG helped to enhance the factual correctness and to decrease the hallucinations.
- The chatbot had structured and easy-to-follow responses all the time.
- The conversational interface was also more user friendly than the traditional search systems.

F. Limitations

Irrespective of its efficiency, the system has some limitations:

- It is not suitable for Professional medical diagnosis.
- The system is based on a predefined query classification logic.
- It is reliant on external APIs, including WhatsApp (Twilio) and Google Places.
- The current implementation does not include user-specific personalization or history tracking.

TABLE II. EXAMPLE USER QUERIES AND CHATBOT RESPONSES

User Query	Retrieved Context Source	Chatbot Response Summary
What are the symptoms of dengue fever?	FAISS-retrieved medical document	Provides structured list of symptoms including high fever, headache, joint pain, and rash, along with guidance to seek medical attention if symptoms worsen.
How can I prevent malaria?	Preventive healthcare knowledge base	Suggests preventive measures such as mosquito control, use of bed nets, and maintaining hygiene, based on retrieved medical guidelines.
What is paracetamol used for?	Drug information corpus	Explains usage of paracetamol for fever reduction and pain relief, including dosage precautions and safe usage recommendations.
What should I do if I have persistent cough and fever?	Symptom analysis knowledge base	Identifies possible causes and provides general guidance while recommending consultation with a healthcare professional.
Where can I find a nearby hospital?	Location-based hospital finder tool (API)	Requests user location and provides nearby healthcare facilities with relevant details using geolocation services.

TABLE III. PERFORMANCE EVALUATION METRICS

Metric	Value	Description
Routing Accuracy	71%	Correct module selection from queries
Response Relevance	91%	Responses match user intent
Safety Compliance	97%	Emergency detection and disclaimers
Average Response Time	3.7 sec	End-to-end latency
Hospital Detection Success	80%	Location-based hospital results

TABLE IV. COMPARISON WITH TRADITIONAL SEARCH SYSTEMS

Evaluation Factor	Traditional Search Engine	Proposed Chatbot System
Information Access	Requires browsing multiple sources	Provides direct summarized answers
Response Structure	List of web links	Structured conversational responses
Ease of Understanding	Often technical and detailed	Simplified and user-friendly explanations
Context Awareness	Limited (keyword-based retrieval)	Context-aware responses using RAG
Risk of Misinterpretation	High due to multiple and sometimes unrelated results	Low due to controlled and focused responses
Emergency Handling	Not available	Integrated emergency detection and guidance
Accessibility	Requires web navigation skills	Accessible via WhatsApp interface
User Effort	High (multiple steps and link navigation)	Low (single conversational interaction)
Response Time	Moderate (20–60 seconds)	Fast (2–4 seconds)

VI. Conclusion And Future Work

In this paper, as described an AI-based public health chatbot intended to enhance the opportunity to get adequate disease awareness and preventive health care information. The suggested system combines Retrieval-Augmented Generation (RAG) and semantic document retrieval with the help of a FAISS vector database so that the responses could be based on verified medical sources of knowledge. The system enhances the possibility of contextually relevant healthcare advice by integrating document-based retrieval with the generation of large language models to minimise the threat of hallucinated responses. The modular design has integrated the medical knowledge retrieval, symptom awareness guidance, drug information retrieval tools and hospital location discovery tools. Experimental analysis showed that the chatbot can retrieve the necessary medical information and provides clear and structured answers without violating the safety restrictions that do not allow diagnostic or treatment prescriptions. Connection to the WhatsApp messaging platform also increases the accessibility of the tool, as people in rural and semi-urban areas can learn about health care services using communication means that they are familiar with.

Even though the current system shows encouraging outcomes, there are multiple opportunities that can be used to improve the current system in the future. The chatbot can be expanded with multi-lingual support that will allow speaking regional languages and make it more accessible to various groups of users. Also, SMS-based deployment has the potential to enlarge the coverage of regions with less internet connections or smartphone accessibility. Additional applications in the future can be automated knowledge base updates through incorporation of trusted public health databases in order to keep the medical information updated. Moreover, the chatbot could adopt an active health notification and prevention strategy by providing real-time health information and predictive analytics on the possibility of upcoming outbreaks to users, with advanced outbreak surveillance systems. These improvements would also help make conversational AI systems more supportive of scalable public health awareness efforts.

References

[1] L. Laranjo, A. G. Dunn, H. L. Tong, A. B. Kocaballi, J. Chen, R. Bashir, D. Surian, B. Gallego, F. Magrabi, A. Y. S. Lau, and E. Coiera, "Conversational agents in healthcare: A systematic review," *Journal of the American Medical Informatics Association*, vol. 25, no. 9, pp. 1248–1258, 2018.

[2] T. W. Bickmore, H. Trinh, R. Asadi, and S. Olafsson, "Safety first: Conversational agents for health information," *Journal of Biomedical Informatics*, vol. 82, pp. 1–9, 2018.

[3] L. Wilson and M. Marasoiu, "The development and use of chatbots in public health: Scoping review," *JMIR Human Factors*, vol. 9, no. 4, p. e35882, 2022.

[4] F. Branda, M. Stella, C. Ceccarelli, F. Cabitza, G. Ceccarelli, A. Maruotti, M. Ciccozzi, and F. Scarpa, "The role of AI-based chatbots in public health emergencies: A narrative review," *Future Internet*, vol. 17, no. 4, p. 145, 2025.

[5] K. Singhal et al., "Large language models encode clinical knowledge," *Nature*, vol. 620, pp. 172–180, 2023.

[6] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung, "Survey of hallucination in natural language generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 1–38, 2023.

[7] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 9459–9474, 2020.

[8] G. Izacard and E. Grave, "Leveraging passage retrieval with generative models for open-domain question answering," in *Proc. 16th Conf. European Chapter Association for Computational Linguistics (EACL)*, pp. 874–880, 2021.

[9] A. S. Miner, L. Laranjo, and A. B. Kocaballi, "Chatbots in the fight against the COVID-19 pandemic," *npj Digital Medicine*, vol. 3, no. 65, pp. 1–4, 2020.

[10] A. Devaraj, A. Sharma, and P. Ghosh, "Evidence-based clinical decision support using natural language processing and retrieval systems," *Journal of Biomedical Informatics*, vol. 128, p. 104002, 2022.